



What is the physical item?

The physical items are realistic clay models of 14 selected fungal specimens, including species native to Singapore as well as commonly known edible fungi consumed worldwide. These specimens belong to the two largest fungal phyla: Ascomycota (sac fungi) and Basidiomycota (club fungi).

The models are accompanied by a handout that explains the differences between these two groups in terms of their characteristics, as well as providing introductions, interesting facts, and the ecological roles they play in the environment, using various examples of fungi.

This activity is designed to be self-exploratory. Members of the public will reach into a cardboard box and randomly pick out a clay fungus model. Based on their observations, they will attempt to classify the fungus as either Ascomycota or Basidiomycota. They may also refer to the handout for more detailed information.

What are you trying to educate others about fungi through your assignment?

By allowing the public to explore various fungi on their own through this self-directed learning approach, I hope to pique their interest and encourage curiosity about both native fungi and those commonly found in Singapore.

The public may already be familiar with edible mushrooms such as truffles, morels, oyster mushrooms, and bamboo fungi. By including their common names, participants will be able to recognise and relate to them more easily.

In particular, I aim to broaden their understanding that Singapore's humid tropical climate supports both Basidiomycota (club fungi) and Ascomycota (sac fungi), which are the two largest fungal phyla. Many of these fungi can be found right here in Singapore. Through this activity, the public will be exposed to a wide variety of species with different shapes and sizes, enabling them to better appreciate the local biodiversity.

Background on Fungal Phyla

Ascomycota is the largest fungal phylum, comprising more than half of all described fungi, with about 64,000 species (Lee et al., 2025). Members of this group can reproduce both sexually and asexually.

Asexual spores, known as conidia, are borne on specialised stalks called conidiophores (Watkinson et al., 2015). Sexual spores, known as ascospores, are formed within sac-like structures called asci (National Parks Board, 2020).

The fruiting bodies of Ascomycota, known as ascomata, occur in three main forms:

- Apothecia (cup-shaped)
- Perithecia (flask-shaped with a single opening for spore release)
- Cleistothecia (closed structures with various mechanisms for spore release)

Lichens are symbiotic relationships between fungi (usually Ascomycota) and algal or cyanobacterial partners (Watkinson et al., 2015). This group also includes edible fungi such as truffles, as well as yeasts used in the fermentation of bread and beverages (Lee et al., 2025).

Basidiomycota is the second-largest fungal phylum, comprising about one-quarter of all described fungi, with approximately 32,000 species (Lee et al., 2025).

In this group, spores known as basidiospores are formed externally on club-shaped structures called basidia (National Parks Board, 2020). Within the basidia, meiosis produces four haploid nuclei, which develop into basidiospores (Watkinson et al., 2015).

Clamp connections are present in many Basidiomycota species. These structures help maintain the dikaryotic state by regulating the distribution of nuclei between hyphal cells.

Basidiomycota includes many familiar fungi, such as mushrooms with prominent fruiting bodies, jelly fungi, and parasitic forms such as rust fungi found on plant leaves (Lee et al., 2025; Watkinson et al., 2015). It also includes bracket fungi, which are typically long-lasting, as well as more fragile, short-lived species such as the Bridal Veil Stinkhorn.

Many Basidiomycota species play important ecological roles in forest ecosystems. They form mycorrhizal symbiotic relationships with trees and are essential for wood decomposition (Watkinson et al., 2015).

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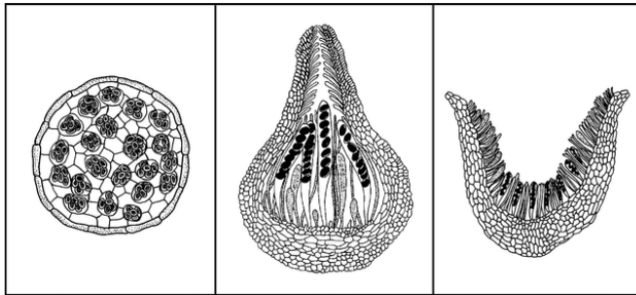
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Exploring Ascomycota and Basidiomycota

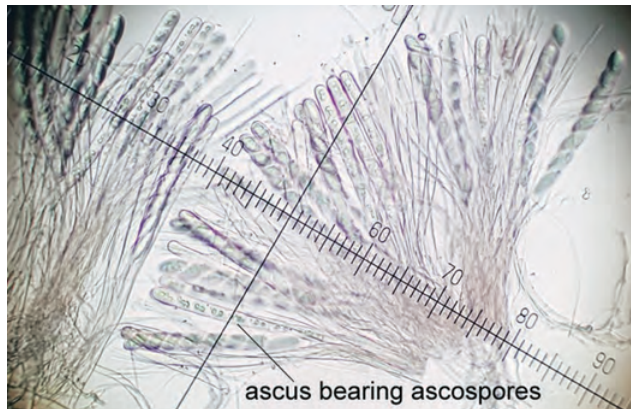
Ascomycota (Sac Fungi)

- Produces ascospores inside sac-like structures called asci
- Fruiting bodies called ascomata



Cleistothecia Perithecia Apothecia

- Includes cup fungi, truffles, morels, yeasts
- Many are decomposers, symbionts, or parasites



1. *Cookeina speciosa*

- Native to Singapore
- Decomposes wood
- The cup-shaped fruiting body allows rain droplets to accumulate, helping to disperse spores
- Spores produced inside asci lining the cup surface



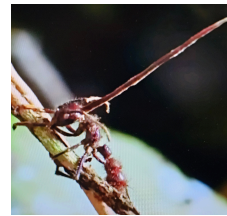
2. *Trichaleurina javanica*

- Peanut-butter Cup Fungus
- Cup-shaped fruiting body
- Decomposer



3. *Ophiocordyceps* sp.

- Zombie Ant Fungus
- Parasite
- The fungus infects and manipulates ants, causing them to climb to a suitable location before it grows and releases spores
- Produces Perithecia



4. *Xylaria* sp.

- Dead Man's Fingers, as the name suggests, are finger-like black structures
- Native to Singapore
- Decomposes wood
- Contains embedded Perithecia



5. *Daldinia eschscholtzii*

- Coal Fungus
- Native to Singapore
- Round, charcoal-like appearance
- Decomposer



6. *Morchella* sp.

- Morel Mushroom (Edible)
- Honeycomb-like cap
- Can form mycorrhizal symbiotic relationships with trees, but they can also feed on decaying organic matter
- Produces spores in sac-like asci



7. *Tuber melanosporum*

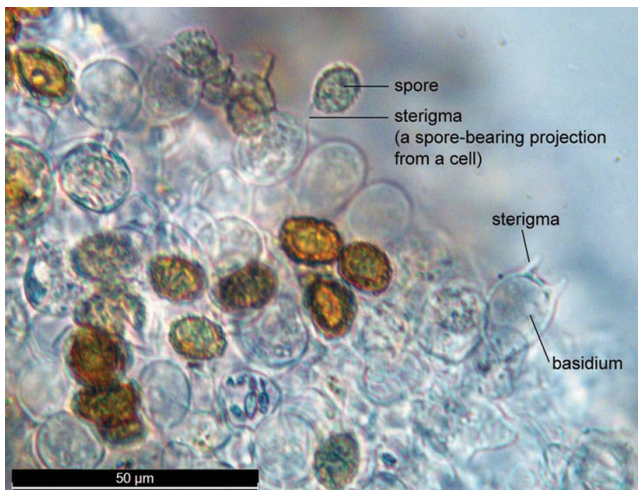
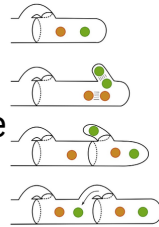
- Black Truffle (Edible)
- Grows underground
- Forms mycorrhizal symbiotic relationships with trees
- Produces spores in asci within a highly modified Apothecia



Exploring Ascomycota and Basidiomycota

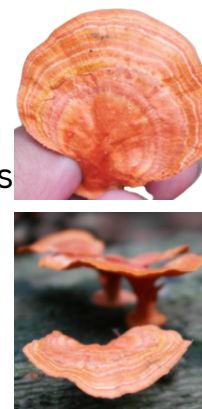
Basidiomycota (Club Fungi)

- Produces basidiospores on club-shaped structures called basidia
- Majority undergo Ballistospory, the forcible ejection of spores
- Most also have clamp connections on their hyphae
- Many form mushrooms, bracket fungi, puffballs
- Important decomposers and symbionts
- Some are edible or plant pathogens



1. *Pycnoporus sanguineus*

- Cinnabar Bracket
- Native to Singapore
- Bright orange bracket fungus featuring pores on the underside, commonly found on dead wood and logs
- Decomposes wood
- By breaking down lignin in dead wood, it helps sustain forest ecosystems



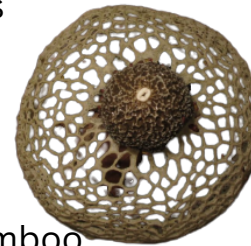
2. *Pleurotus djamor*

- Pink Oyster Mushroom (Edible)
- Featuring gills on the underside, it decomposes wood, recycling nutrients back into the soil
- Found on fallen logs



3. *Phallus indusiatus*

- Bridal Veil Stinkhorn/ Bamboo Fungus (Edible)
- Native to Singapore
- Easily recognisable by its lacy veil
- Decomposer
- Unique among Basidiomycota, it emits a strong odor to attract insects, such as flies, for spore dispersal



4. *Ganoderma lucidum*

- Lingzhi/ Reishi (Edible)
- Native to Singapore
- Used in Traditional Chinese Medicine (TCM)
- Infect living trees and persist in breaking down wood even after the tree has died



5. *Cyathus striatus*

- Fluted Bird's Nest Fungus
- Resembles small bird's nest with "eggs"
- Native to Singapore
- Decomposer
- Spore dispersal occurs via rain, as raindrops splash and release the spores



6. *Geastrum sp.*

- Earthstar
- Star-shaped fruiting body consisting of an outer layer surrounding an inner spore sac, which features a central opening
- Decomposer



7. *Scleroderma verrucosum*

- Poison Puffballs
- Native to Singapore
- Spherical fruiting bodies whose outer layer ruptures at maturity to release spores
- Decomposer

